

Building a Premier League Prediction Challenge App with Python and Streamlit

Abstract

This report details the development of an interactive Premier League Prediction Challenge App using Python and Streamlit. The app allows participants to submit their predictions for the final league standings, tracks real-time standings, and calculates scores based on the accuracy of predictions. The aim was to create a fun and competitive way for football enthusiasts to engage with the Premier League season.

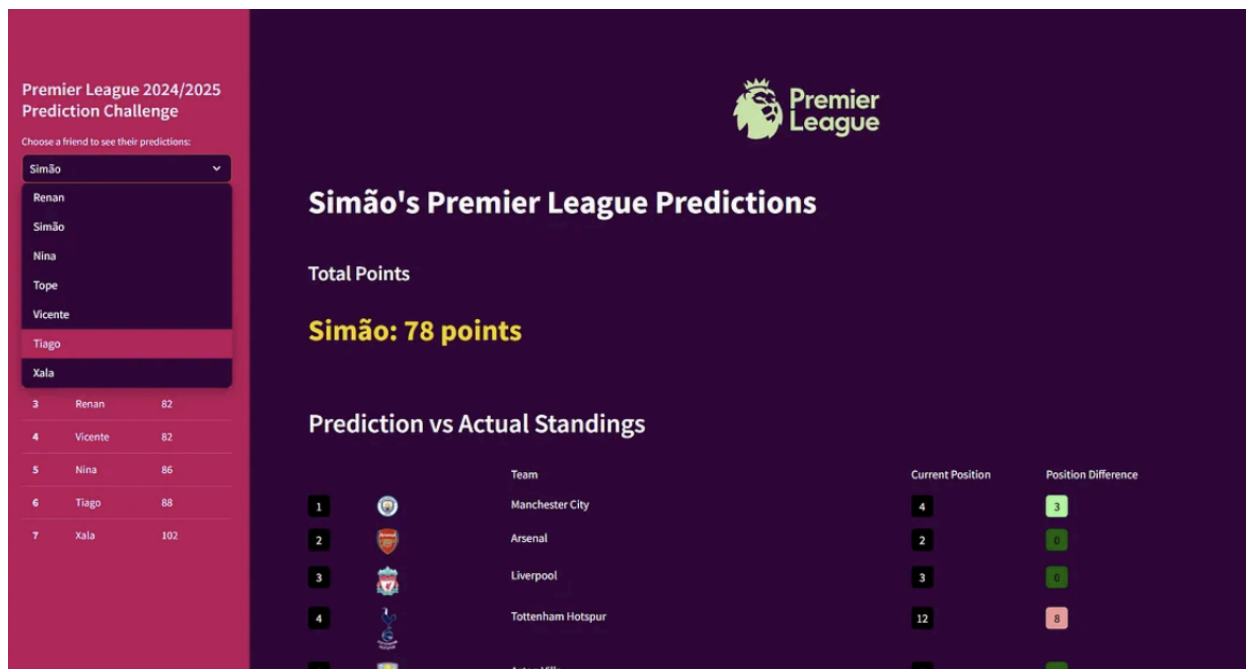


Figure 1 | App interface

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1. Introduction

The Premier League is a popular football league, and an app can be created to test fan's football knowledge by predicting league standings. The problem is to develop an application that allows users to predict final Premier League standings and compare their predictions against actual results.

The objectives are to build an app for participants to submit Premier League standings predictions, track actual Premier League standings in real-time, calculate and display scores based on prediction accuracy, and provide an interactive and user-friendly interface to view predictions and standings.

The contributions will be an interactive app that enhances user engagement with the Premier League, a practical application of Python and Streamlit for a real-time prediction challenge, and a visually appealing and intuitive interface for users to track their predictions.

2. Approach

The app was built using Python for core functionality and Streamlit for the user interface and deployment. It involves data extraction, data processing, and comparison of predictions with actual standings.

2.1 Data Extraction

Algorithm/Technique Used: Web scraping using BeautifulSoup and requests to extract live Premier League standings from the BBC Sports website.

Implementation Details: The requests library fetches the HTML content from the BBC Sports website. BeautifulSoup parses the HTML to find the table containing the league standings. The data is then extracted and transformed into a Pandas DataFrame.

2.2 Data Processing and Comparison

Algorithm/Technique Used: Pandas DataFrames for data manipulation and comparison. The app compares the prediction DataFrames submitted by participants with the live Premier League standings.

Implementation Details: Participants' predictions are submitted as CSV files, which are read into Pandas DataFrames. The app calculates points based on the difference between the predicted position and the actual final position for each team. The scoring system rewards fewer points for more accurate predictions.

2.3 User Interface and Deployment

Algorithm/Technique Used: Streamlit for creating an interactive web interface and deploying the app online.

Implementation Details: Streamlit is used to create a user-friendly interface where participants can view their predictions, track the live standings, and see how well they are doing. Custom HTML and CSS are integrated to enhance the visual presentation, including color-coded formatting to indicate prediction accuracy.

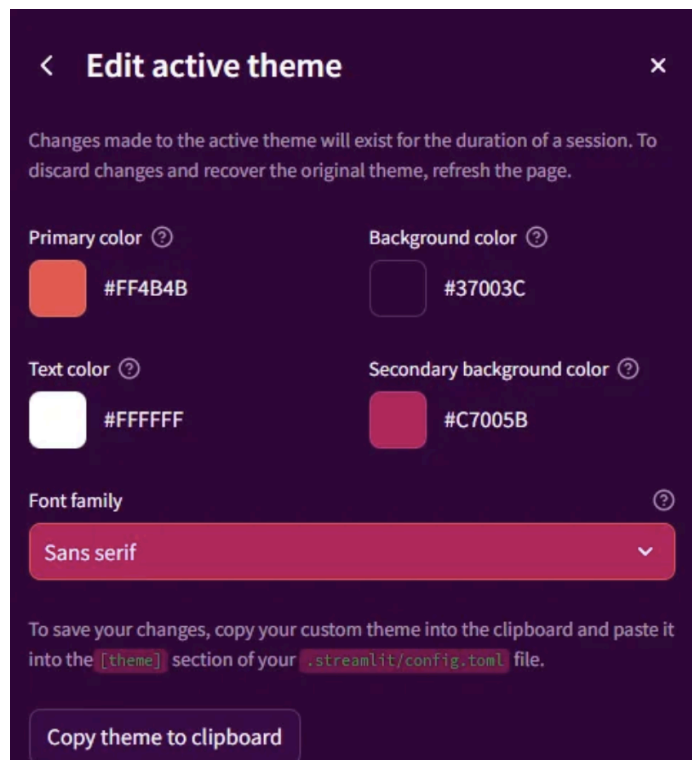


Figure 2 | Streamlit custom theme

3. Experiment

The data used for this analysis comes from live Premier League standings on the BBC Sports website. To measure the accuracy of predictions, we compare them against the actual final standings. The evaluation metric is based on points calculated from the difference between predicted and actual positions; the lowest score wins.

The app merges prediction DataFrames with the current standings DataFrame to find the difference in positions. This data is presented in a table with team logos, predicted positions, current positions, and the difference between them. The accuracy of each prediction is color-coded, ranging from green (very accurate) to red (significantly off).

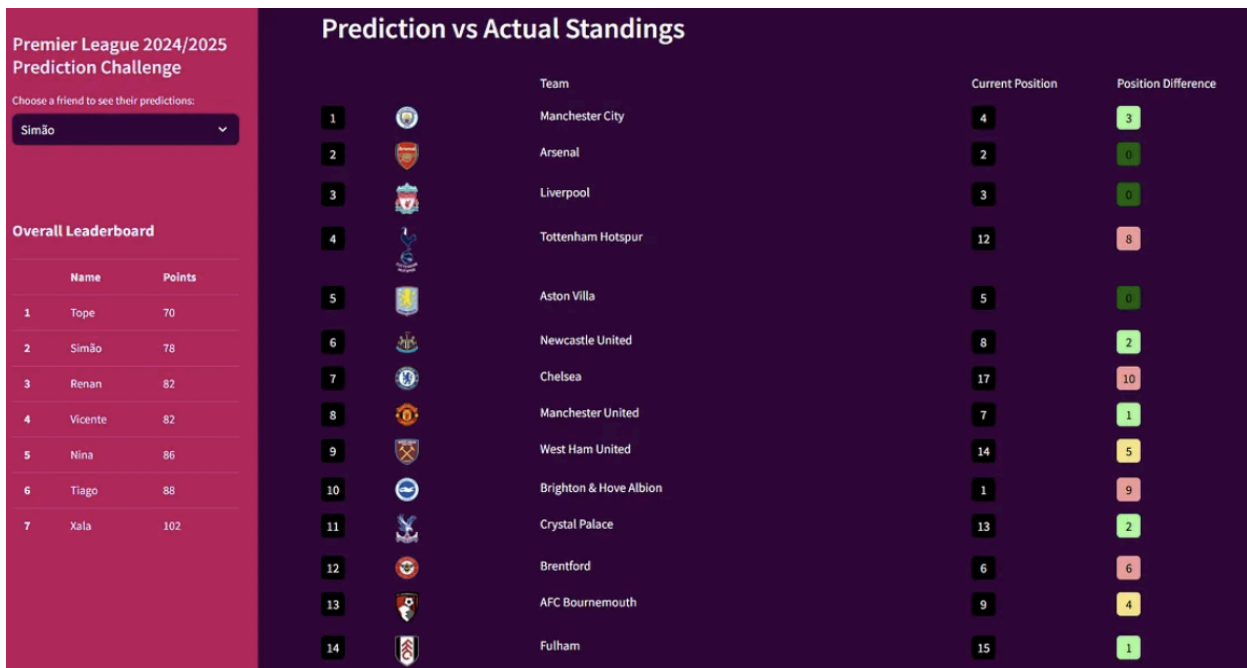


Figure 3 | App Leaderboard

4. Discussion

The analysis of the results shows that the app successfully provides an engaging and interactive platform for users to track and compare their Premier League predictions. The real-time data updates, achieved through web scraping, ensure that users have access to the most current information, which enhances the overall user experience. Additionally, the integration of custom HTML and CSS allows for a visually appealing and polished interface that distinguishes the app from other prediction platforms. The app's use of color-coded formatting and visual enhancements further contributes to its user-friendly design, enabling users to quickly and easily assess the accuracy of their predictions.

5. Conclusion, Limitations, and Future Work

Limitations:

- The app is not mobile-friendly.
- The accuracy of real time predictions depends on the quality of data from the BBC Sports website.

Future Work:

- Improve mobile responsiveness for better accessibility.

- Implement user authentication and database storage for predictions.
- Incorporate more advanced scoring metrics and prediction analytics.